Engineering Portfolio

Athan Stefanos

Email: athanstefanos@gmail.com

Mobile: 0419503041

Introduction

My name is Athan and I am an Aerospace engineering student.

Welcome to my portfolio, where I go through the engineering projects I have worked on over the years since I started university.

I love hands-on problem solving and challenging myself with new experiences. I see this as a fundamental aspect of living, and I hope it shows through my work.

Thank you for your time and enjoy the portfolio!

Project | 2025
Meridian

Manufacturing & Integration
AURC 2026

POSITION
Flight Systems Engineer

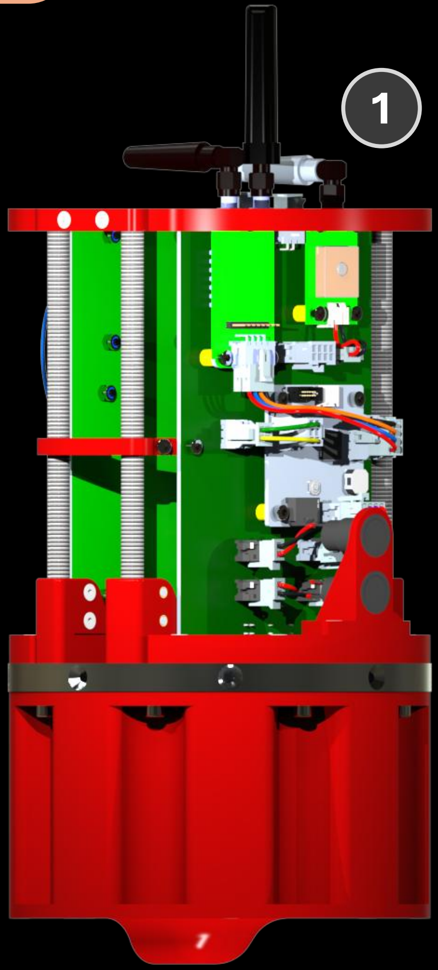
The design of an electronics bay for a sounding rocket that will compete in the Australian University Rocket Competition in the 10,000 ft category.

Contribution – as of 02/26

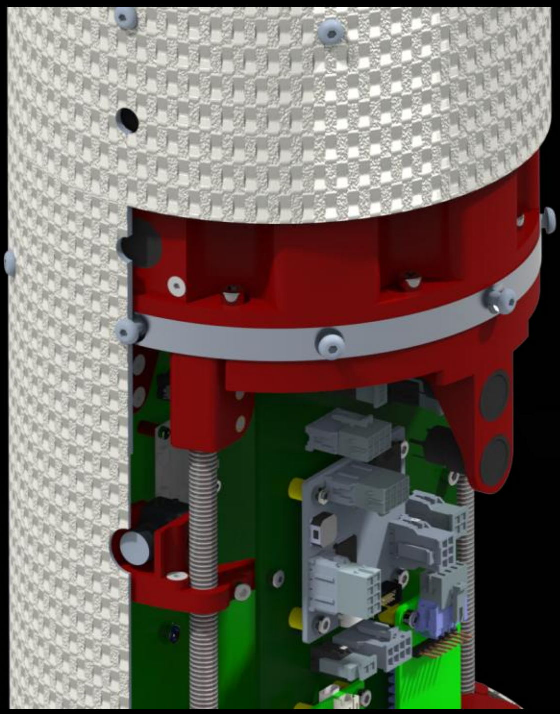
- Complex design of a structure to **interface** with 3 individual sections. [1]
- **Rapid prototyping** of different designs to accommodate **stakeholder feedback** [2].
- Designing for manufacturing and **high-force** environments. [3]

Learnings

- Learnt how to interface with multiple stakeholders simultaneously and effectively gather system requirements
- Learnt the importance of understanding surrounding systems and effectively recommended improvements
- Furthered skill in complex CAD modelling and assembly (Creo Parametric, SolidWorks)



1



2

3

Project | 2025
Stick Bug

POSITION
CAD Designer, Stability Analyst, Vice-lead

The design of a commercial jet to enter the middle of the aviation market with the goals of sustainability and cost-effectiveness in mind.

Contribution

- **Management** of a large team and the **facilitation** of multiple **technical and economic requirements**. [1]
- **Stakeholder analysis** and management of a **full design cycle**.
- Aircraft **stability analysis** and tailoring of plane configuration to meet **mission requirements. [2]**

Learnings

- Skills gained in **aircraft performance analysis** and design principles for commercial aircraft.
- Professional presentation and graphics for a business proposal. [3]

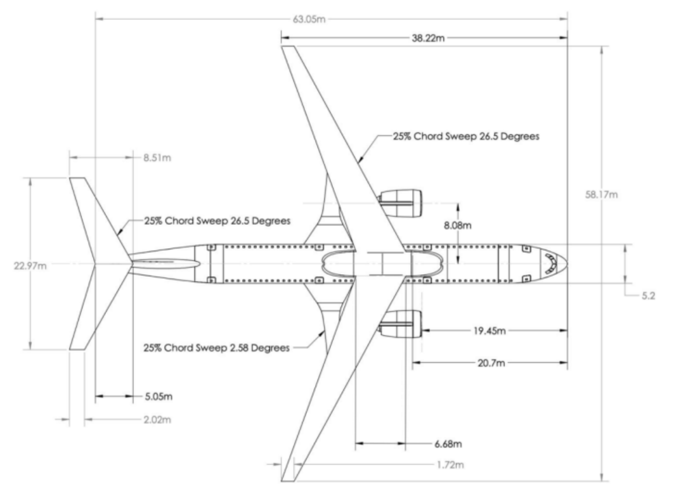
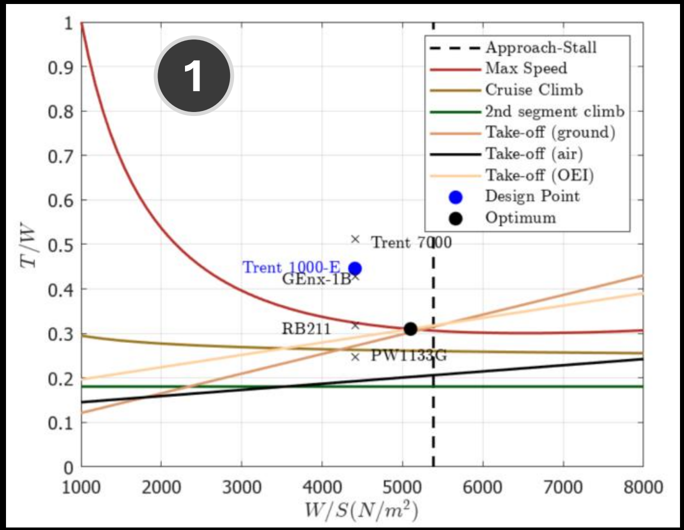
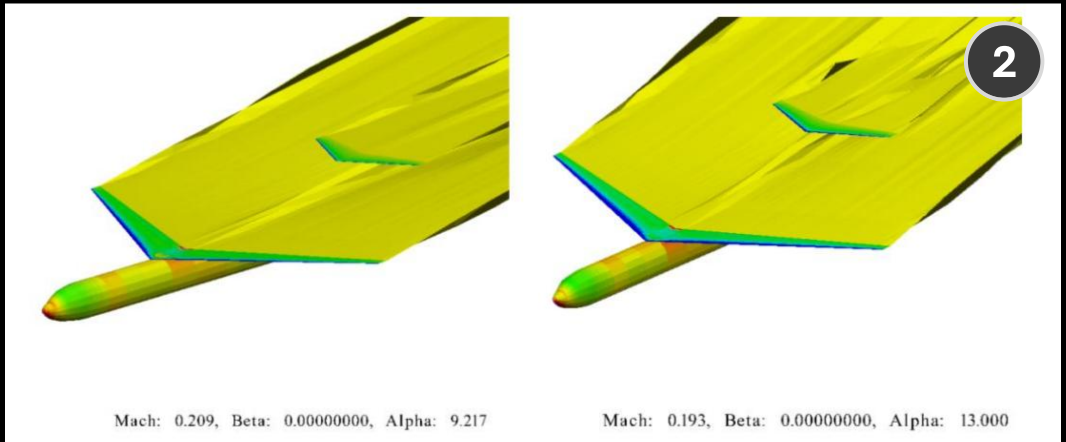
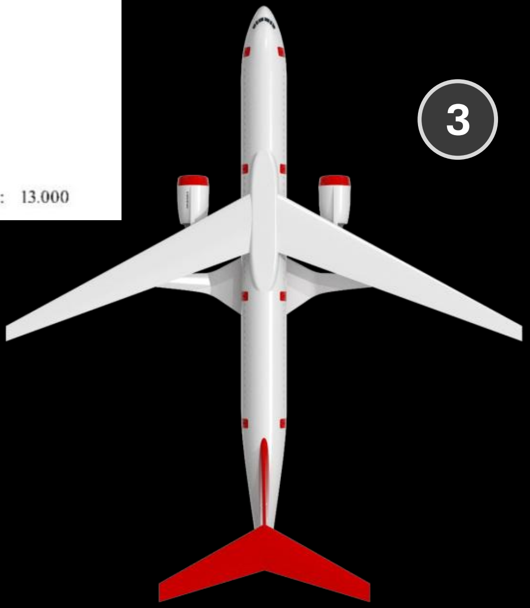


Figure 0-3: Stick Bug Top View







Project | 2025
Zenith

Successfully Launched
IREC 2025

POSITION
Aerostructures Engineer

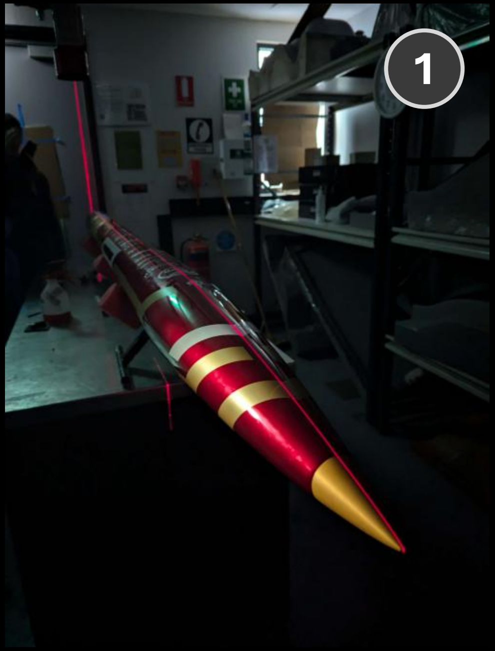
The design of a 10,000 ft sounding rocket airframe to compete in the International Rocket Engineering Competition.

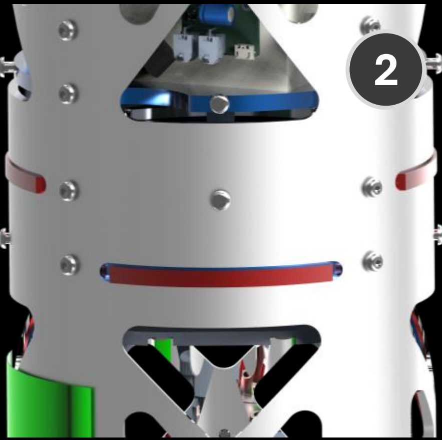
Contribution

- **Design, validation** and **integration** of aluminium chassis to house air brakes and electronic systems [2].
- **Manufacturing** and **assembly** of a **composite** fibreglass/carbon fibre 3 metre rocket. [1]

Learnings

- **Finite Element Analysis** of **isotropic** and **anisotropic** rocket components (**ANSYS**).
- Fibreglass and carbon fibre composite manufacturing techniques (**wet layup**, **resin infusion**).
- Manufacturing and **integration** problem solving.
- Spray painting and prep work.





Project | 2025
RC Plane

POSITION
Co-lead,
Aerodynamics,
powerplant

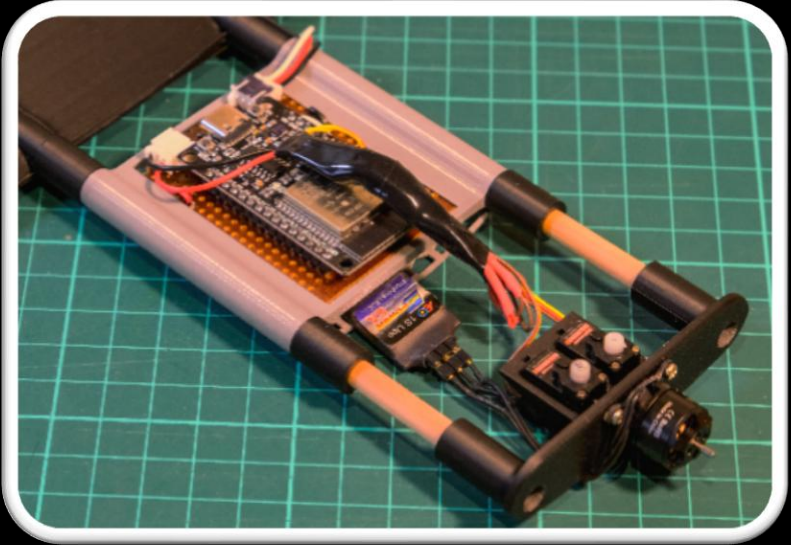
This project was the design of a custom RC plane platform to aid in the teaching of flight vehicle dynamics to VCE and first year university students.

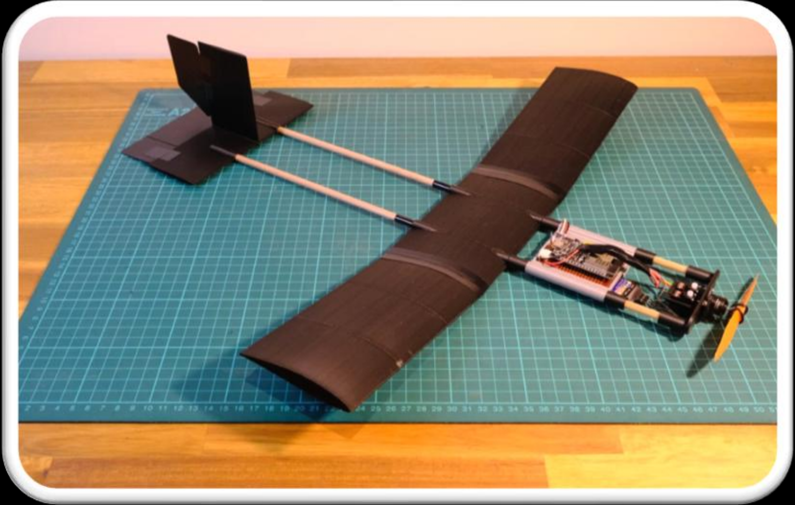
Contribution

- **Management** of a tasks and deliverables within a small team
- **Stakeholder analysis** and market studies to determine viability and use.
- Aircraft **stability analysis** and tailoring of plane configuration to meet **mission requirements.**
- **Rapid prototyping** and flight tests to verify designs. [1]

Learnings

- Additive manufacturing refinement and optimisation
- Application of flight vehicle performance theory
- Group management





Project | 2024 Wyvern

Successfully Launched
AURC 2025

POSITION
Aerostructures Engineer

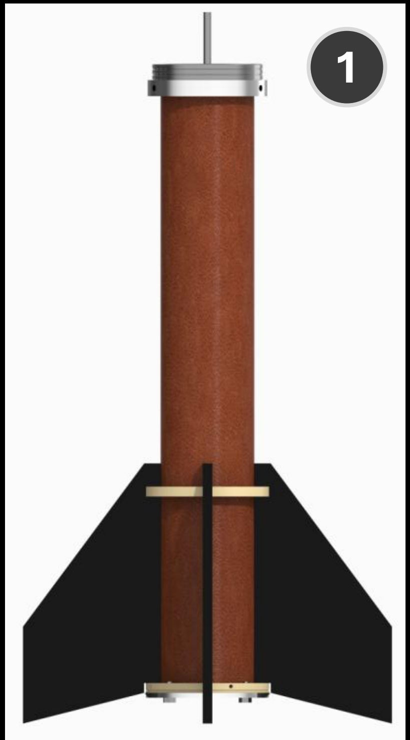
The design of a sounding rocket tail section to reach 5000 ft that has competed in the Australian Universities Rocket Competition.

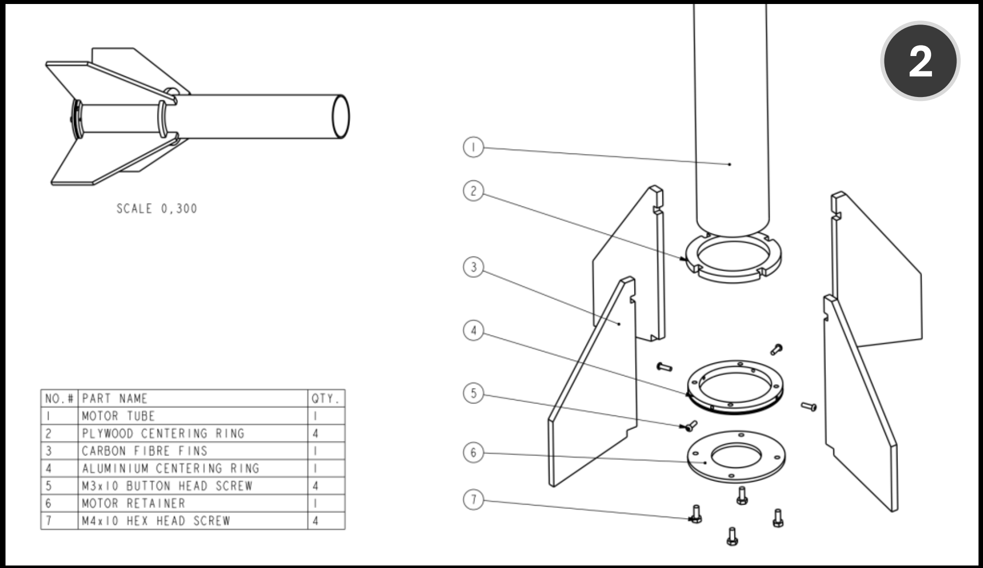
Contribution

- **Design, validation, Manufacturing** and **assembly** of a rocket empennage (motor tube, fincan). [1]
- Solution **cost analysis** and **project management**.
- Part/assembly technical drawings (SolidWorks). [2]

Learnings

- **Managing team members** within a project **timeline** for component design, manufacturing and assembly
- **Interdisciplinary communication** and **teamwork**.
- **Aerodynamic analysis** (fin flutter).





NO.#	PART NAME	QTY.
1	MOTOR TUBE	1
2	PLYWOOD CENTERING RING	4
3	CARBON FIBRE FINS	1
4	ALUMINIUM CENTERING RING	1
5	M3x10 BUTTON HEAD SCREW	4
6	MOTOR RETAINER	1
7	M4x10 HEX HEAD SCREW	4

Ballute – Rocket Recovery

The critical design of a novel drogue parachute for a 100,000 ft sounding rocket.

- Investigation of requirements and conditions of a project outside of the team's experience.
- In-depth literature review to source potential solutions and designs.
- Generation of suitable geometry, material selection, and manufacturing techniques tailored to the intended use of the product.